

Economic Structural Resilience in OECD Countries

A POSet-Based Framework for Multi-Dimensional Economic Comparison

Colombini Francesco - 944941 Haider Ali - 948092
Università degli Studi di Milano Bicocca Data Science Lab Project
{f.colombini4, a.haider16}@campus.unimib.it

KEYWORDS Economic Resilience · Partial Order Theory · Pareto Dominance · Macroeconomic Panel Data

ABSTRACT Traditional resilience frameworks aggregate heterogeneous economic indicators into single composite scores, introducing arbitrary weighting and forced linear rankings. This project proposes a Partial Order Theory (POSet) framework to analyse structural resilience across OECD countries without constructing a composite index. Countries are compared through Pareto dominance relations across five pre-shock structural indicators: debt capacity, employment strength, R&D intensity, tertiary human capital and energy security. The methodology separates structural ordering variables from crisis validation variables, using the Global Financial Crisis (2007 baseline) and COVID-19 (2019 baseline) as validation episodes. The resulting structures show high incomparability (0.68 in 2007, 0.77 in 2019). Frontier countries recover faster after both shocks and outperform non-frontier countries on five out of six post-shock macroeconomic indicators in both periods.

AI Tools Declaration. Generative AI tools (Claude by Anthropic and ChatGPT by OpenAI) were employed exclusively for code refinement and debugging during the implementation of the data acquisition, preprocessing and POSet construction pipeline. All research design decisions, methodological choices, analytical interpretations, and written content were produced independently by the authors.

1. Introduction

This paper investigates whether pre-shock economic structures are associated with stronger resilience outcomes across OECD countries during systemic crises. The central question is not which country is “best”, but whether some structural configurations place countries in a stronger position before a shock occurs.

The project uses Partial Order Theory (POSet) because economic resilience is multidimensional [12]. A country can be strong in innovation and human capital but weaker in debt capacity or energy security, while another country may show the opposite pattern. Reducing these different configurations to a single composite score would hide these trade-offs and force comparisons that may not be meaningful.

For this reason, the analysis does not construct a universal Economic Resilience Index. Instead, countries are represented through five pre-shock

structural variables: debt capacity, employment strength, R&D intensity, tertiary human capital and energy security. These variables are selected because they describe relatively stable economic capacities rather than short-term crisis outcomes.

The POSet is constructed separately for two structural baselines: 2007, before the Global Financial Crisis, and 2019, before the COVID-19 shock. The resulting partial orders are then validated using post-shock macroeconomic outcomes, including GDP recovery, unemployment, inflation, debt and productivity. This separation between ordering variables and validation variables is essential: the POSet measures structural capacity before the shock, while validation checks whether stronger structural positions are associated with better outcomes after the shock.

A key contribution of the framework is that it treats incomparability as an informative result. If two countries cannot be ordered because each

has strengths in different dimensions, the model reports this directly rather than hiding the difference inside a weighted average. The framework therefore provides a structural map of economic resilience configurations, identifying frontier countries, dominated countries and incomparable profiles without forcing a complete linear ranking.

2. Research Objective

The main research question is:

Which structural economic configurations are associated with stronger resilience during systemic crises?

Answering this question requires addressing a prior issue that many resilience scorecards leave implicit. Composite indicators often reduce several dimensions to one weighted value and therefore assume that every country can be placed on a single line. This assumption is problematic for resilience analysis because countries may rely on different structural strengths [9]. One economy may have lower debt and stronger energy security, while another may have stronger innovation capacity and human capital. A single score would make these cases appear directly comparable even when their structural profiles are different.

The research question therefore has two parts. First, to what extent are OECD economies structurally comparable to one another? Second, among the countries that are comparable, do structurally dominant profiles show stronger post-shock outcomes? Partial Order Theory is suited to this double question because it tests for dominance relations without assuming that all countries must be comparable [5]. Incomparability is therefore not treated as a modelling error, but as evidence that different economic structures may represent different pathways to resilience.

This framing is useful for policy interpretation. If one country structurally dominates another, it becomes reasonable to ask whether some features of the stronger profile can inform reform priorities. If two countries are incomparable, the framework signals that policy transfer should be interpreted more carefully because each country may have strengths in different dimensions. The purpose is not to produce a league table, but to support a more transparent comparison of structural resilience profiles. We address this double question in three steps: first constructing the structural

partial order from pre-shock indicators, then verifying its robustness to alternative dominance thresholds, and finally testing whether structurally dominant positions are associated with stronger post-shock outcomes.

3. Data Acquisition & Preprocessing Pipeline

Answering the research question requires a dataset that separates what an economy looks like structurally before a shock from how it performs during one. The project adopts a reproducible data acquisition pipeline implemented in Python. All datasets are collected and merged into a unified country-year panel indexed by country and year. The workflow integrates OECD Data Explorer [1], Eurostat [2] and World Bank [3] sources.

Most macroeconomic variables are retrieved from OECD SDMX endpoints. These include R&D expenditure, tertiary education attainment, GDP growth, unemployment, productivity and inflation-related indicators. Public debt is handled through a source-priority rule: Eurostat is preferred where available, while the World Bank public-debt series is used as a fallback. This avoids keeping two overlapping public-debt values for the same country-year pair and creates a single canonical public-debt variable.

Energy import dependency is retrieved from the World Bank and retained as an external vulnerability measure. Since higher raw energy dependency indicates greater exposure, the variable is direction-aligned into an energy security score before entering the POSet.

The Worldwide Governance Indicators (WGI) are collected and processed as contextual information, but they are not used as a final ordering variable. This is a deliberate methodological choice. WGI is already a composite governance dataset built from several dimensions and underlying sources. Since the project criticizes hidden aggregation and forced composite rankings, including WGI as a main ordering variable would partly reintroduce a pre-aggregated composite inside the POSet itself. Governance remains relevant, but it is treated as contextual or sensitivity material rather than as one of the five baseline ordering dimensions.

The final dataset is divided into two subsets. The first subset contains the structural ordering variables used to construct the POSet. The sec-

Table 1: Data Sources Inventory

Dataset	Source	Identifier	Coverage	Role
OECD_RD_GDP	OECD SDMX	DSD_MSTI@DF_MSTI · UNIT=PT_B1GQ, MEASURE=G	2002– 2025	Ordering
OECD_Tertiary_Educ.	OECD SDMX	DSD_EAG_LSO_EA@DF_LSO_NEAC_DISTR_ EA · Y25T64, ISCED11A_5T8	2002– 2025	Ordering
OECD_Unemployment	OECD SDMX	DSD_LFS@DF_LFS_INDIC · MEASURE=UNE_RATE, PT_LF_SUB	2002– 2025	Ordering / Validation
OECD_GDP_Growth	OECD SDMX	DSD_NAMAIN1@DF_QNA_EXPENDITURE_GR OWTH_OECD · B1GQ.PC.GY	2002– 2025	Validation
OECD_Inflation_CPI	OECD SDMX	DSD_PRICES@DF_PRICES_ALL · CPI._T.GY	2002– 2025	Validation
OECD_Productivity	OECD SDMX	DSD_PDB@DF_PDB_LV · GDPHRS.USD_PPP_H.V	2002– 2024	Validation
Eurostat_Public_Debt	Eurostat	Public debt/GDP series from Eurostat REST dissemination API	2002– 2024	Ordering / Validation
WB_Public_Debt	World Bank	GC.DOD.TOTL.GD.ZS	2002– 2025	Fallback
WB_Energy_Import	World Bank	EG.IMP.CON.S.ZS	2002– 2025	Ordering
WGI (6 dimensions)	World Bank	WGI bulk CSV — databank.worldbank.org/data/d ownload/WGI_CSV.zip	1996– 2024	Context

ond subset contains validation variables used only for post-shock evaluation. This separation prevents outcome leakage: the variables used to judge crisis performance are not allowed to influence the structural ordering.

3.1. Strengths and Weaknesses of the Data

The main strength of the dataset is its institutional provenance. All variables are drawn from official OECD, Eurostat or World Bank sources. This improves reproducibility because each variable can be retrieved again through documented APIs or public bulk downloads. It also makes the definitions externally auditable rather than specific to this project.

A second strength is the clear separation between structural variables and validation variables. The ordering variables describe pre-shock capacities, while the validation variables measure post-shock outcomes. This design is important because it allows the project to test whether the structural POSet has explanatory value without building the outcome directly into the ordering.

The main limitation is coverage. Complete-case requirements reduce the sample to 25 countries for the 2007 baseline and 35 countries for the 2019 baseline. This is preferable to filling structural gaps with artificial values, but it means the final POSet represents only countries with sufficient data coverage. A second limitation is that some potentially relevant dimensions, such as governance quality, industrial structure or supply-

chain exposure, are not included in the final baseline POSet. In particular, WGI is not used as an ordering variable because it is already a composite measure; however, it remains useful for interpretation and possible future extensions.

3.2. Ordering Variables

The final POSet is constructed using five structural variables. Each variable is direction-aligned so that a higher score indicates a more favourable structural condition.

Table 2: POSet Ordering Variables

Variable	Interpretation	Direction
Debt capacity	Lower fiscal fragility	↑
Employment strength	Lower labour-market stress	↑
R&D intensity	Innovation capability	↑
Tertiary human capital	Adaptive skills base	↑
Energy security	Lower external energy exposure	↑

The five variables are selected because they are slow-moving structural indicators: each one reflects the underlying configuration of an economy over the medium-to-long term rather than its cyclical position at any given point in time. This distinction matters for the internal validity of the analysis. If the ordering variables captured cyclical dynamics, the resulting POSet would conflate structural capacity with short-term fluctua-

tions, making it impossible to isolate whether pre-existing structural conditions drive resilience outcomes, or whether countries simply entered the crisis in a favourable cyclical position. Variables that are conceptually relevant but exhibit higher cyclical sensitivity, such as GDP growth, were assigned to the validation set rather than the ordering set on these grounds.

Debt capacity is derived from public debt as a share of GDP, with lower debt interpreted as greater fiscal space. Employment strength is derived from the unemployment rate, with lower unemployment interpreted as a more favourable labour-market structure. R&D intensity measures innovation capacity, while tertiary human capital captures the stock of adult skills and adaptability. Energy security is derived from energy import dependency and captures exposure to external supply conditions. These five variables are intentionally simple and directly interpretable, which keeps the POSet readable and avoids hidden composite weighting.

3.3. Validation Variables

Validation variables are not used in the construction of the POSet itself. They are used only to evaluate whether stronger pre-shock structural positions are associated with better post-shock macroeconomic outcomes.

Table 3: *Validation Variables*

Variable	Purpose
GDP recovery time	Speed of recovery after shock
Average GDP growth	Post-shock output performance
Average unemployment	Labour-market resilience
Unemployment change	Labour-market deterioration
Absolute inflation	Price stability
Public debt change	Fiscal pressure after shock
Productivity change	Supply-side continuity

The validation indicators are deliberately kept separate from the ordering process. GDP recovery time measures how long it takes a country to return to its pre-shock GDP path. The macro-indicator validation then checks whether frontier countries perform better than non-frontier countries across GDP growth, unemployment, inflation stability, debt dynamics and productivity. The validation is descriptive rather than causal: it tests

whether the structural map is empirically meaningful, but it does not claim that the selected variables alone caused the observed recovery outcomes.

3.4. Data Preprocessing

The preprocessing phase harmonizes heterogeneous international datasets into a common country-year structure. Country identifiers and years are standardized, duplicate country-year-variable observations are checked, and source-specific inconsistencies are resolved before constructing the analytical panel.

Missing data are handled conservatively. The final POSet uses complete-case samples for the selected baseline year and the five ordering variables. Countries with incomplete structural profiles are excluded from the relevant baseline rather than being filled through interpolation. This reduces sample size but avoids creating artificial structural profiles.

All final ordering variables are direction-aligned so that higher values always mean a more favourable condition. Public debt and unemployment are inverted into debt capacity and employment strength, while energy import dependency is inverted into energy security. R&D intensity and tertiary education already have a positive interpretation. The direction-aligned variables are then scaled into comparable 0–100 scores and transformed into five ordinal levels for POSet construction.

The POSet is constructed using single-year structural snapshots: 2007 for the Global Financial Crisis baseline and 2019 for the COVID-19 baseline. This choice reflects the purpose of the analysis: the unit of interest is the structural configuration of an economy immediately before a shock, not its average trajectory over a long period. Medium-term volatility measures are computed only as contextual diagnostics and are not included in the final baseline POSet.

3.5. Tools

The full pipeline is implemented in Python, using modular notebooks for acquisition, preprocessing, POSet construction, sensitivity analysis, validation and visualization. The code is structured so that each stage produces intermediate diagnostic outputs and final report-ready tables. All code procedures are available in the project repository [7].

4. Partial Order Formulation and Structural Profiles

The POSet translates multidimensional country profiles into a structure that does not force comparability where none exists. Each country is represented by five direction-aligned structural variables: debt capacity, employment strength, R&D intensity, tertiary human capital and energy security. A country is considered structurally dominant only when it is at least as strong as another country in every dimension and strictly stronger in at least one dimension.

This rule is non-compensatory: a country cannot compensate for weakness in debt capacity, employment strength or energy security simply by having a high value in R&D or tertiary education. This is the key difference between the POSet and a composite index, keeping the separate dimensions visible and reporting when two countries cannot be compared without introducing arbitrary weights [10].

4.1. Temporal Scope of Structural Measurement

Each ordering variable is measured at a single pre-shock year: 2007 for the Global Financial Crisis baseline and 2019 for the COVID-19 baseline. This design keeps the structural ordering temporally prior to the crisis outcomes used for validation. It also makes the interpretation simple: the framework asks whether the structure observed before a shock is associated with later resilience outcomes.

The validation windows are separate from the ordering years. For the 2007 baseline, validation uses post-shock macroeconomic behaviour from 2008–2012. For the 2019 baseline, validation uses the available post-COVID period from 2020–2023. This separation protects the analysis from outcome-year leakage.

A multi-year average of the ordering variables could be used in future work to smooth short-term anomalies. In this project, however, the single-year snapshots are chosen to preserve temporal priority and interpretability. Because the selected variables are relatively slow-moving structural indicators [12], the risk of short-term volatility is lower than for high-frequency macro outcomes. Nevertheless, multi-year baseline averaging remains a robustness extension.

4.2. Ordinal Profiling

The continuous structural scores are converted into five ordinal levels within each shock-specific baseline sample. The five ordinal levels are constructed using within-sample quantile-based discretization for each shock baseline and each variable. Level 1 represents the weakest group in a dimension and level 5 represents the strongest one. The five dimension levels are then combined into a structural profile code. For example, a profile such as 3-5-5-2-2 describes a country that is relatively strong in some dimensions and weaker in others. Ordinal levels are interpreted within each baseline year and are not intended as absolute cross-year measurement units.

This profiling step reduces the influence of small numerical differences. A very small difference in public debt or unemployment should not necessarily imply a different structural type. By using ordinal levels, the POSet focuses on broader structural configurations rather than minor decimal-level variation.

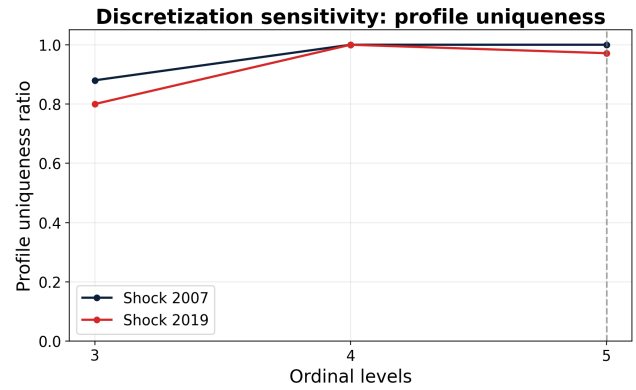


Figure 1: Discretization sensitivity of the profile construction. The figure compares profile uniqueness across alternative numbers of ordinal levels and supports the use of five levels as the main specification.

Five levels are retained because they preserve sufficient profile differentiation without producing a fully unique profile for every country in all cases. Lower levels collapse too many countries into broad groups, while higher levels risk making the POSet overly granular and less interpretable.

4.3. Dominance Rule and Epsilon-Margin Robustness

The main profile POSet uses strict Pareto dominance on the ordinal profiles. A profile dominates another profile if it is at least as good in all five dimensions and better in at least one dimension. This produces the main Hasse diagrams and the

main frontier results.

As a robustness check, the analysis also applies an ϵ -margin rule on normalized country-level scores. Under this rule, dominance requires not only that one country is at least as good in every dimension, but also that it has a minimum advantage in at least one dimension. This avoids interpreting negligible numerical differences as meaningful structural superiority. The epsilon-margin analysis is a robustness check, not a replacement for the main profile-level POSet.

4.4. Graph Reduction and Structural Incomparability

The dominance relations between profiles form a directed acyclic graph. A transitive reduction is then applied to remove redundant indirect relations and produce a Hasse diagram. In the diagram, an edge represents a direct dominance relation between two profiles.

The absence of a path between two profiles means these profiles are structurally incomparable: each may have strengths in different dimensions, and neither can be said to dominate the other without introducing subjective trade-offs. High incomparability shows why a single ranking would hide important structural information.

4.5. Empirical POSet Results

The empirical POSet was constructed separately for the 2007 and 2019 baselines. The final specification uses five ordering variables and five ordinal levels. The 2007 sample contains 25 countries and 25 distinct profiles. The 2019 sample contains 35 countries and 34 distinct profiles. In both years, the incomparability ratio is high, which confirms that OECD economies cannot be adequately described by a complete ranking.

Table 4: Summary of Final POSet Structures

Metric	GFC 2007	COVID 2019
Countries included	25	35
Distinct profiles	25	34
Pareto-frontier countries	8	13
Pareto-frontier share	32.00%	37.14%
Dominance relations	96	129
Direct Hasse edges	52	64
Comparability ratio	0.3200	0.2299
Incomparability ratio	0.6800	0.7701
POSet layers	5	4

For the 2007 baseline, eight countries are located on the Pareto frontier. For the 2019 baseline, thirteen countries are located on the Pareto

frontier. A frontier country is not necessarily the best country in every dimension; rather, it is not dominated by another observed profile. This distinction is important because the POSet identifies structural non-domination without converting it into a universal ranking.

Country	Public Debt	Employment Strength	R&D intensity	Tertiary Human Capital	Energy Import Independency
AUT	2	4	5	3	2
BEL	1	2	4	4	1
CAN	3	3	4	5	5
CZE	4	3	3	1	5
DEU	2	1	4	2	3
DNK	4	5	5	4	5
ESP	4	1	3	3	1
EST	5	5	2	5	5
FIN	3	2	5	5	3
FRA	2	2	4	3	3
GBR	1	4	3	5	5
GRC	1	1	1	2	2
HUN	2	2	2	1	3
IRL	4	4	2	4	1
ITA	1	3	2	1	2
LTU	5	5	1	3	3
LUX	5	5	3	3	1
LVA	5	3	1	2	2
NLD	3	5	4	4	4
POL	3	1	1	2	4
PRT	1	2	2	1	1
SVK	4	1	1	1	2
SVN	5	4	3	2	4
SWE	3	3	5	4	4
USA	2	4	5	5	4

Table 5: Ordinal profile scores for the country comparison in 2007.

The POSet does not rank countries from strongest to weakest. Instead, it separates countries into comparable and incomparable structural configurations. This is particularly important for resilience analysis because different economies may reach strong structural positions through different combinations of debt capacity, employment strength, innovation, human capital and energy security. Dominance relations include all pairwise dominance links, while direct Hasse edges remain after transitive reduction removes redundant indirect paths.

Profile POSet Hasse diagram

2007 baseline: Global Financial Crisis
Node labels show ordinal profile code and ISO country code(s)

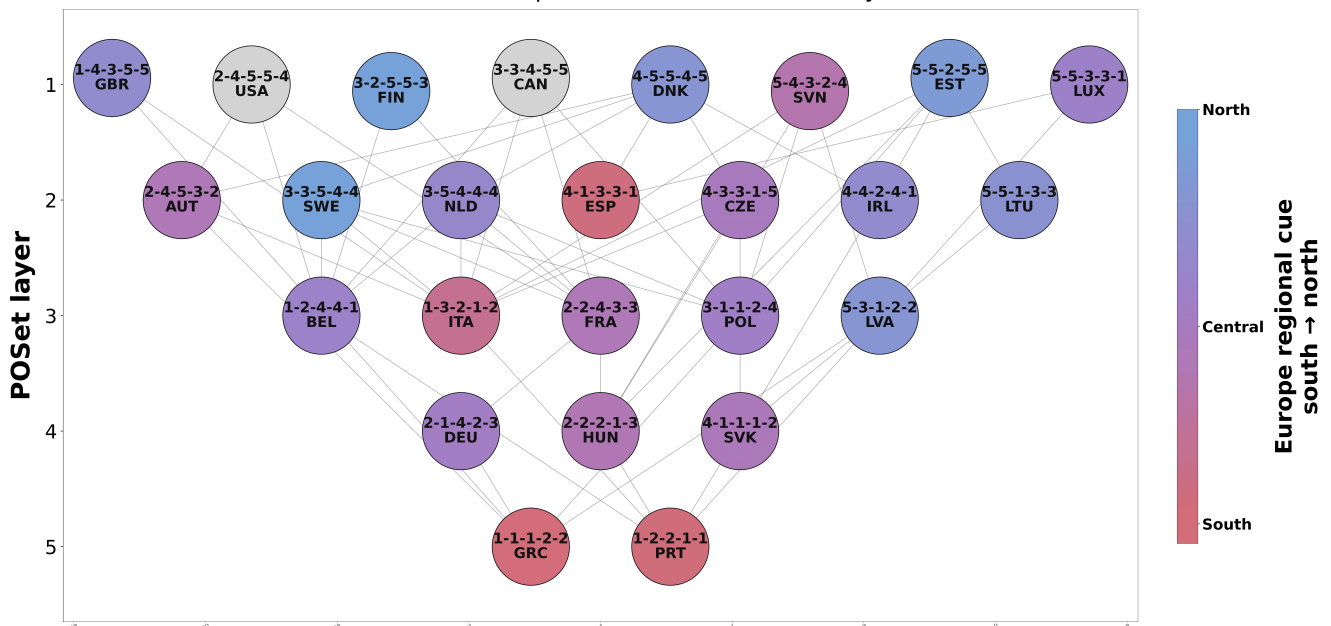


Figure 2: Reduced Hasse diagram for the 2019 baseline of COVID-19. Each node represents a five-variable structural profile, and each edge represents a direct dominance relation after transitive reduction.

Country	Public Debt	Employment Strength	R&D intensity	Tertiary Human Capital	Energy Import Independency
AUS	3	3	3	5	5
AUT	2	3	5	3	2
BEL	1	3	5	4	1
BGR	5	4	1	2	4
CAN	3	3	3	5	5
CHE	5	4	5	4	3
COL	2	1	1	1	5
CZE	5	5	4	1	4
DEU	3	5	5	2	2
DNK	4	3	4	3	4
ESP	2	1	2	3	1
EST	5	4	3	4	5
FIN	2	2	4	4	4
FRA	1	1	4	3	3
GBR	1	5	4	5	4
GRC	1	1	2	2	1
HRV	2	2	2	1	3
HUN	2	5	3	2	2
IRL	3	4	2	5	2
ITA	1	1	3	1	1
KOR	4	5	5	5	1
LTU	4	2	1	4	2
LUX	5	3	2	5	1
LVA	4	2	1	3	3
NLD	3	4	4	4	2
NZL	5	4	3	3	4
POL	4	5	2	2	3
PRT	1	2	3	2	1
ROU	4	4	1	1	5
SVK	3	2	1	2	3
SVN	2	3	4	3	3
SWE	4	2	5	4	4
TUR	5	1	2	1	2
USA	1	5	5	5	5
ZAF	3	1	1	1	5

Table 6: Ordinal profile scores for the country comparison in 2019.

4.6. Robustness of the Partial Order

The robustness analysis tests whether the main POSet conclusions depend on small numerical differences. The epsilon-margin rule was tested over values from 0.00 to 0.20 on normalized country-level scores with steps of 0.05. In both shock baselines, all tested epsilon-margin values remained valid partial orders, with no reciprocal dominance cycles.

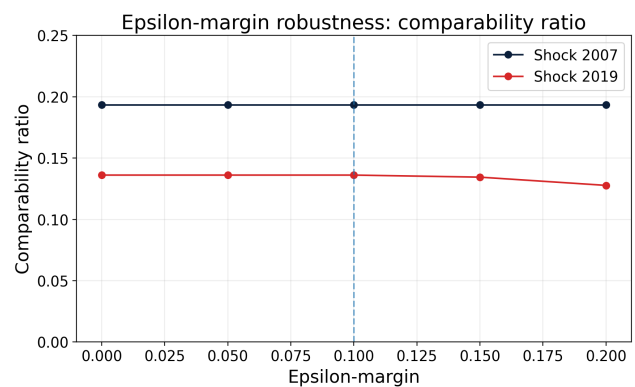


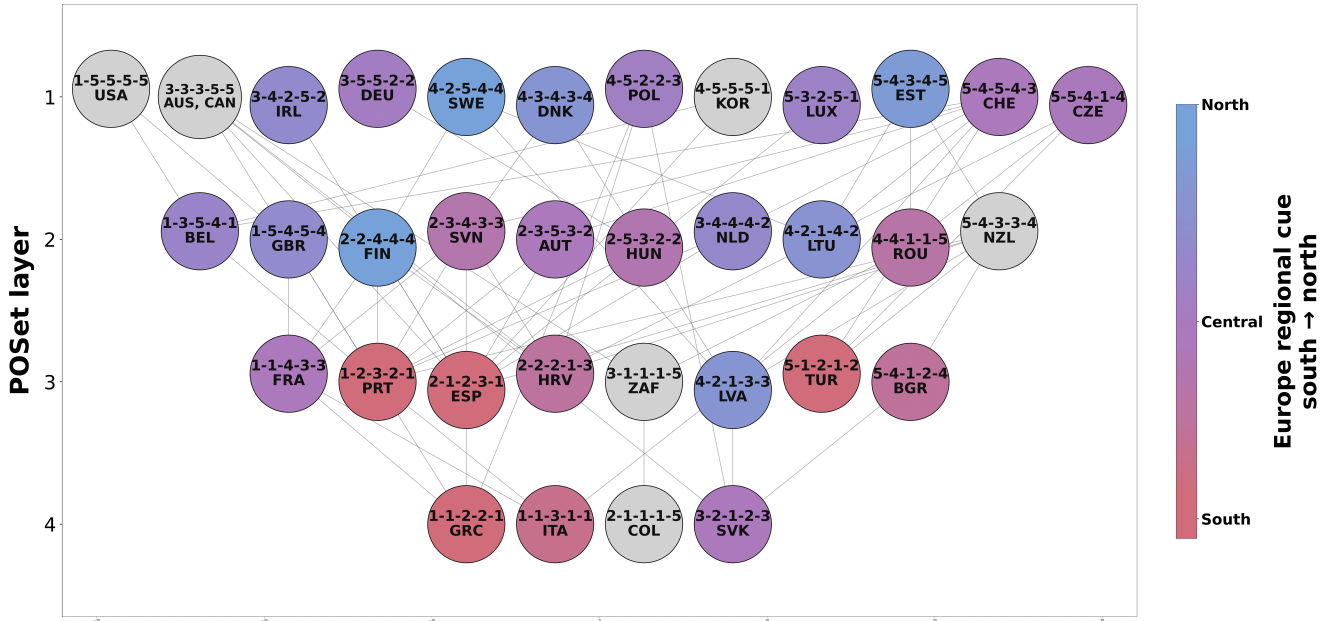
Figure 4: Epsilon-margin robustness of the country-level dominance relation. The figure shows how the comparability ratio changes across tested epsilon-margin values while preserving partial-order validity.

The conclusions are not driven solely by

Profile POSet Hasse diagram

2019 baseline: COVID-19 shock

Node labels show ordinal profile code and ISO country code(s)



Vertical position shows POSet layer. Horizontal position is layout only and has no economic interpretation.

Figure 3: Reduced Hasse diagram for the 2019 COVID-19 baseline. The diagram shows both dominance relations and the large number of structurally incomparable profiles.

marginal numerical differences. The epsilon-margin analysis is not used as the main ranking tool. It is used to confirm that the dominance logic remains coherent under stricter dominance requirements.

5. Shock Validation

The objective of the validation stage is to assess whether the pre-shock structural ordering contains meaningful information about resilience during systemic crises. As discussed above, validation uses independent post-shock outcomes that are different from the variables used in the POSet construction. This ensures that the validation procedure tests the explanatory value of the structural hierarchy rather than reproducing information already embedded in it.

5.1. Validation Design

The selected variables are hypothesized to capture resilience-related capacities, while countries in stronger POSet positions should show more favourable outcomes after a systemic shock. The analysis evaluates this idea in two ways.

First, GDP recovery validation compares the average recovery time of frontier and non-frontier countries. Recovery time measures how long it takes an economy to return to its pre-shock GDP

level or reconstructed baseline index. Second, multi-indicator validation compares frontier and non-frontier countries across six macroeconomic outcomes: average GDP growth, average unemployment, unemployment change, absolute inflation, public debt change and productivity change. This comparison is used as an interpretable summary of structural non-domination. It does not exhaust the POSet structure, which also contains layer and incomparability information.

The validation windows are 2008–2012 for the Global Financial Crisis baseline and 2020–2023 for the COVID-19 baseline. These variables are not used to construct the POSet. They are used only after the structural profiles have been defined. These validation indicators are direction-aligned before comparison, so positive frontier advantage consistently means better average post-shock performance. Also, the windows are chosen to capture the immediate and medium-term post-shock adjustment period while avoiding excessive contamination from later unrelated macroeconomic cycles.

5.2. GDP Recovery Validation

The validation results are consistent with the structural interpretation of the POSet. In the 2007 baseline, frontier countries recover faster on aver-

age by approximately 0.82 years compared with non-frontier countries. In the 2019 baseline, frontier countries recover faster on average by approximately 0.26 years. Because of the limited sample size, the validation is interpreted descriptively. The results indicate directional alignment suggesting that structurally undominated profiles are associated with stronger recovery outcomes. It must be considered that the frontier groups are relatively small, especially in 2007 (8 countries vs 13 in 2019), so average differences may be influenced by individual country cases and outliers.

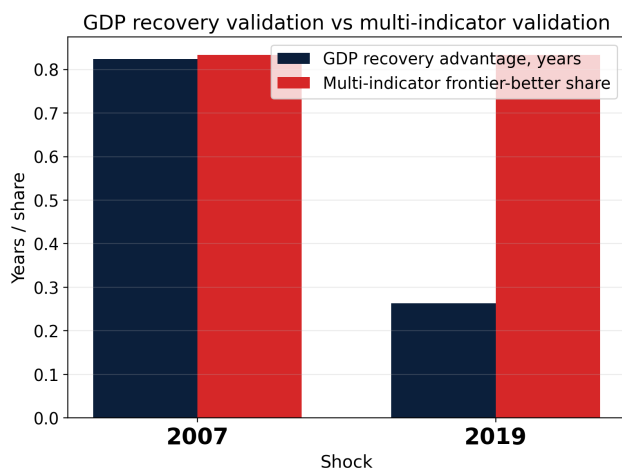


Figure 5: Comparison between GDP recovery validation and broader multi-indicator validation. Frontier countries show faster recovery on average and outperform non-frontier countries on most macroeconomic validation indicators.

5.3. Multi-Indicator Validation

The multi-indicator validation provides a broader test of the POSet structure. For both the 2007 and 2019 shocks, frontier countries outperform non-frontier countries on five out of six available macroeconomic validation indicators. In 2007, the only contradictory indicator is productivity change, which favours non-frontier countries. In 2019, the only contradictory indicator is unemployment change, which also favours non-frontier countries. These contradictions are not treated as errors; they indicate that resilience outcomes are multidimensional and can be affected by shock-specific policy responses, labour-market institutions and sectoral exposure.

Table 7: Final Validation Summary

Metric	GFC 2007	COVID 2019
Countries included	25	35
Frontier countries	8	13
GDP recovery frontier advantage	0.82 years	0.26 years
Macro indicators favouring frontier	5/6	5/6
Main contradictory indicator	Productivity change	Unemployment change
Interpretation	Supportive	Supportive

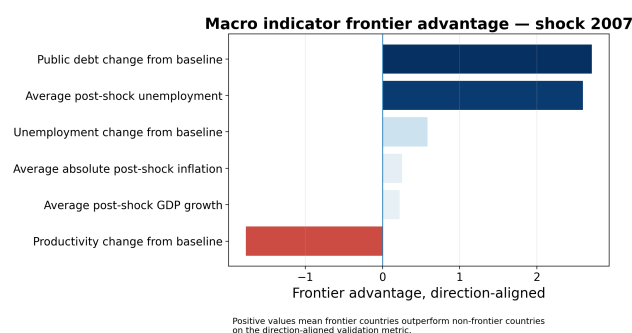


Figure 6: Direction-aligned frontier advantage by macroeconomic validation indicator for the 2007 Global Financial Crisis baseline. Positive values indicate that frontier countries perform better than non-frontier countries on the corresponding post-shock outcome.

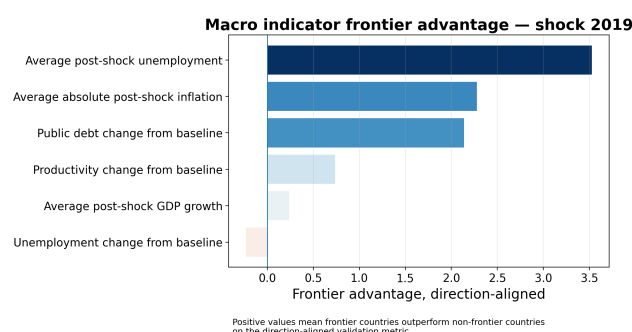


Figure 7: Direction-aligned frontier advantage by macroeconomic validation indicator for the 2019 COVID-19 baseline. Positive values indicate that frontier countries perform better than non-frontier countries on the corresponding post-shock outcome.

The validation evidence is not perfect, and the mismatch cases are important. Some frontier countries underperform on specific outcomes, while some deeper-layer countries perform relatively well. This is expected in a multidimensional framework. Crisis outcomes also depend on policy responses, shock exposure, sectoral composition and international conditions. The validation should therefore be interpreted as evidence of alignment between structural position and later outcomes, not as proof of causality.

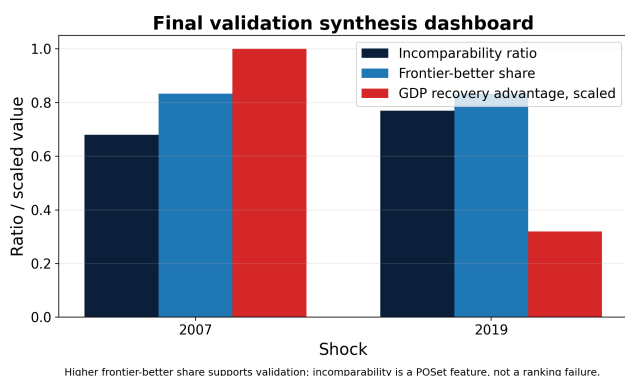


Figure 8: *Final validation synthesis dashboard combining POSet incomparability, multi-indicator validation support and GDP recovery advantage. The figure summarizes the validation evidence without converting the POSet into a scalar index.*

6. Conclusion

This project examined whether pre-shock economic structures are associated with stronger resilience during major crises in OECD countries. Rather than constructing a composite resilience ranking, the analysis used Partial Order Theory to compare countries across five structural dimensions: debt capacity, employment strength, R&D intensity, tertiary human capital and energy security. The framework was validated against two systemic shocks: the 2008 Global Financial Crisis, using a 2007 structural baseline, and the COVID-19 pandemic, using a 2019 structural baseline.

The main finding is that OECD countries cannot be described adequately through a single ranking from strongest to weakest. In both baselines, the majority of country pairs are structurally incomparable: the incomparability ratio is 0.68 in 2007 and 0.77 in 2019. This means that countries frequently combine structural strengths and weaknesses in different ways, and that forcing a linear ranking onto these configurations would hide meaningful structural information rather than summarise it.

The validation evidence is supportive but not uniform, in fact frontier countries recover faster on average after both shocks, by approximately 0.82 years in the GFC baseline and 0.26 years in the COVID-19 baseline. They also outperform non-frontier countries on five out of six post-shock macroeconomic indicators in both periods. The two contradictory cases are interpretable rather than anomalous. In 2007, productivity change favours non-frontier countries because frontier economies tend to have more financially integrated productive structures, making them more directly exposed to a shock that originated pre-

cisely in the financial system. In 2019, unemployment change favours non-frontier countries because economies with already weaker pre-shock labour markets had fewer employed workers exposed to lockdown disruption, producing smaller measured unemployment variation during the crisis. These asymmetries are consistent with the POSet logic: structural strength in one dimension does not guarantee uniform outperformance across every post-shock outcome.

It is also important to acknowledge that treating debt, unemployment and energy dependency as structural liabilities, and innovation capacity and human capital as structural assets, embeds a normative position about what constitutes economic resilience. Alternative frameworks prioritising different dimensions, such as social cohesion, industrial diversification or supply-chain independence, would produce different structural orderings. This project does not claim its value system is the only valid one; it claims only that the chosen dimensions are transparently defined, slow-moving and empirically meaningful.

The framework should therefore act as a structural map for identifying frontier, dominated and incomparable resilience profiles in a transparent and non-compensatory way. Future work could extend the analysis through multi-year structural baselines, additional shock episodes, and the incorporation of further dimensions such as governance quality, supply-chain exposure or economic complexity.

References

- [1] OECD Data Explorer. Available at: <https://data-explorer.oecd.org/>
- [2] Eurostat Database and API. Available at: <https://ec.europa.eu/eurostat/web/main/data/database>
- [3] World Bank Data and API. Available at: <https://data.worldbank.org/>
- [4] OECD, "Economic resilience: A new set of vulnerability indicators for OECD countries."
- [5] Davey, B. A., Priestley, H. A., *Introduction to Lattices and Order*, Cambridge University Press.
- [6] Worldwide Governance Indicators. Available at: <https://www.worldbank.org/en/publication/worldwide-governance-indicators>

- [7] Github Project Folder. Available at: https://github.com/FRA-0023/Economic_Structural_Resilience_in_OECD_Countries
- [8] Bruggemann, R., Patil, G.P. Ranking and Prioritization for Multi-indicator Systems.
- [9] Bruggemann, R., Carlsen, L. An Improved Estimation of Partial Order Ranking.
- [10] OECD (2008) Handbook on Constructing Composite Indicators: Methodology and User Guide
- [11] Martin, R. (2012) Regional Economic Resilience, Hysteresis and Recessionary Shocks.
- [12] Martin, R., Sunley, P. (2015) On the Notion of Regional Economic Resilience.
- [13] Barro, R.J. (2001) Human Capital and Growth.
- [14] Cherp Jewell (2014) The Concept of Energy Security.